

Chapter 5_ Action Teaching AI to do, Not

The Detective's Dilemma

The agent noticed a concerning pattern: an entire collection of premium wines had not moved from the shelves in months. Following its programming for efficiency and armed with its pricing tools, the agent made what seemed like a logical decision: mark them for clearance. What the agent couldn't understand, despite all its sophisticated tools, was that these wines were meant to sit there, aging to perfection and increasing in value. By the time humans notice, the revenue has already evaporated.

They can process vast amounts of data and execute complex tasks with precision, yet they can also miss context that would be obvious to a junior retail clerk.

More tools, more constraints

Paradox: Giving AI more tools makes it less effective. It is like a worker can become overwhelmed with too many applications and systems. AI agents need carefully curated toolsets to perform at their best.

Why tools matter in the age of AI

A rule of thumb: if a digital tool can be operated by a human via a digital interface, it can also be utilized by an agent, provided proper access and instructions are granted.

It is not about the models; it is all about what tools it can access and how it applies them.

The number of tools an agent should access depends on the complexity of its tasks, much like how a human's productivity can depend on the tools they are expected to use in their job. Half a dozen tools may represent a practical maximum for most AI agents.

More tools would cause hallucination, diminishing returns in task efficiency, and heightened risk of conflicts between functionalities.

Tools are clearly defined capabilities with specific input and output.

To instruct the agent, we can deploy a comprehensive framework for tool definition.

We have to provide 5 essential components:

- **Tool identity:** a unique, descriptive name and clear purpose statement
Instruction: Use the calendar scheduler to set up a meeting
- **Input parameter:** exactly what the tool needs to function
Instruction: as input, you will receive the meeting's date, time, and participation
- **Output specifications:** what the tool returns and in what format.
Instruction: As output, use the calendar scheduler to confirm the meeting details once scheduled.

- **Operational constraints:** any limitations or requirements.
Instruction: Use the calendar scheduler only if the selected time slot is available.
- **Error handling:** expected failure modes and recovery procedures.

The building block

- **APIs:** it is a universal translator
- **System controls:** these allow the agent to interact with the software in the way that humans would. In short, it is like giving the model the mouse and the keyboard to operate.

Emergent abilities

It is a capabilities that was not explicitly programmed but arise from their training. It is the capability to break down complex goals into logical steps and understand cause-and-effect relationships.

While the traditional automated system needs explicit programming for every possible scenario, an LLM-powered agent can understand new situations and figure out appropriate tools to use based on its general understanding of how tools work.

When facing a problem, the agent demonstrated an impressive ability to understand its constraints

Then it is the ability to reallocate resource underlining its strength in recalibrating plans.

By the time of speaking in 2026, we are in phase 3 of Agentic AI, it is a "junior stage" where agents are at the stage of problem-solving capability.

Conflicting goals

"Decision paralysis" = the agent either gets stuck trying to satisfy all conditions simultaneously or oscillates between different solutions without concluding.

"False resolution" = they propose compromises that appear to satisfy all requirements but actually subtly violate one or more constraints. For example, they may suggest splitting the document into chunks or smaller files to meet the size requirement while technically violating the security protocol that requires all information to remain in a single encrypted file.

Solution

We don't have to create a new sophisticated model, but we use "priority matrix," which is a human-designed framework that pre-defines how to handle specific types of conflicts. It transforms the model from decision-maker to decision-implementer, playing to its strengths while protecting against its limitations.

We have to think of the agent as "literal-minded assistants" = they can execute complex tasks brilliantly but need clear, non-conflicting instructions to function effectively.

Monitoring and Auditing

Limitation "Context Gap" = AI agent can follow security rules but struggles to understand the broader security implications of their actions. Studies found that the level 3 agents could maintain perfect compliance with security protocols, but they consistently missed security implications that would be obvious to a human security professional.

- Security protocols must be explicitly defined because AI systems cannot intuitively understand or develop security measures on their own.
- Human oversight is indispensable
- Regular security auditing of AI agents is necessary to ensure that they continue to perform as intended and maintain security standards over time.